Reputation Burning: Analyzing the Impact of Brand Sponsorship on Social Influencers

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ABSTRACT

The growth of the influencer marketing industry warrants an empirical examination of the effect of posting sponsored videos on influencers' reputations. We collected a novel dataset of user-generated YouTube videos created by prominent English-speaking influencers in the beauty and style category. We extracted a rich set of theory-driven video features and used DiNardo-Fortin-Lemieux reweighting to construct comparable treatment and control groups matched at the influencer-video level. The empirical analysis of the matched sample revealed a *reputation-burning effect*; that is, posting a sponsored video, compared with posting an equivalent organic video, cost the influencers 0.19% of their reputation (operationalized as the number of subscribers). The reputation-burning effect was stronger among the influencers with larger audiences. An analysis of likes, comments, and comment texts revealed a larger gap in audience response between sponsored and organic videos among the influencers with larger (vs. smaller) audiences. The reputation-burning effect was mitigated when a high fit existed between the sponsored content and the influencer's "usual" content and when the promoted brand was less well-known. Our study empirically tested the assumption of several theoretical works. Moreover, it contributes to the literature on influencer marketing and celebrity endorsements, and provides managerial implications for influencers, brands, and social media platforms.

Keywords: influencer marketing; social influencer; brand sponsorship; video analytics

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1. INTRODUCTION

The influencer marketing industry was reportedly worth USD 6.0 billion in 2020, which is projected to increase to USD 84.89 billion by 2028 (a compound annual growth rate of 30.3%; Globe Newswire 2021). Market reports have shown that 75% of brands dedicated a specific budget to influencer marketing to engage consumers and promote brands (Geyser 2021). For example, *The Ordinary* paid TikTok influencers to post videos about the brand's product Peeling Solution and generated more than $1 million in sales in two weeks.[1]

Influencers are effective marketers because they engage large audiences on social media platforms and appear as experts or trustworthy sources of information for their fans. They build reputations by creating authentic, relevant, and interesting *organic* contents (Sokolova and Kefi 2019). Once influencers become sufficiently popular, they can profit from their reputations by collaborating with brands to create *sponsored* content (McQuarrie, Miller, and Phillips 2013). Our study focused on influencers' reputations with their audiences,[2] operationalized as numbers of subscribers. Avery and Israeli (2020) suggested that influencer reputation is a key driver in influencer marketing because reputation is the foundation of an influencer's ability to obtain profitable offers from brands. Eldor (2019) referred to reputation as "the major currency in the influencer and retail world." However, while brands seek collaborations with influencers for their reputations, several prior theoretical works have implicitly or explicitly assumed that sponsorships may hurt the influencer's reputation (Fainmesser and Galeotti 2021; Mitchell 2021), perhaps by causing the audience to question the influencer's authenticity (Forrester 2018). Empirically, the crucial question as to whether and how sponsored content impacts influencer reputation remains unanswered.

Before we describe our research question and data, we first discuss the nature and implications of influencer reputations and brand sponsorships. According to theoretical works in economics, reputation is

[1] https://www.dailymail.co.uk/femail/article-8026573/Teenager-sparks-1-1MILLION-sales-Ordinary-skincare-posting-TikTok.html.

[2] Note that this measure might not be the same as the influencer's reputation with the public. For example, Donald Trump may have a controversial reputation with the public but have a good reputation with his fans; our study focuses on the latter.

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an important asset that can be accumulated, consumed, and restored; that is, reputation can be cyclical (Liu 2011; Liu and Skrzypacz 2014). Liu (2011) characterized reputation dynamics as interactions between a service provider (e.g., a firm or an influencer) and individual agents (e.g., consumers or an audience). The agents observe the service provider's behaviors and form or update beliefs about the provider's "type" (good or bad reputation), which affects the agents' willingness to have subsequent interactions with the service provider. For example, if consumers observe that a firm handled a customer complaint poorly or launched a bad product, then their perception of the firm's reputation will be negatively affected, possibly stopping them from purchasing from the firm if the perceived reputation is very poor.

Similarly, on social media, users observe an influencer's posting behavior (e.g., the quality and frequency of posts and whether posts are brand-sponsored) and form or update their perceptions of the influencer's reputation, which impacts whether users want to interact or continue to interact with the influencer in the future. Users who perceive that an influencer has a good reputation may follow (or continue to follow) in order to "listen to" the influencer, whereas users who perceive that an influencer has a bad reputation may "reject" the influencer by unfollowing them or not following them in the first place. Intuitively, a user's decision to follow or unfollow an influencer reflects the user's willingness to interact with and hear from the influencer in the long run (Morton 2020). Thus, an influencer gains subscribers in a cumulative process similar to reputation building over time (Mitchell 2021), and the influencer's number of subscribers reflects their reputation and ability to influence others. Brands look out for influencers with many subscribers because subscriber counts indicate influencers' power to influence and persuade people (Eldor 2019), which is the brand's motivation for collaborating with influencers (Avery and Israeli 2020).

How does brand sponsorship impact influencer reputation? Specifically, if an influencer's content characteristics remain similar, would their reputation benefit more from posting sponsored content than from posting organic content? On the one hand, collaboration with brands may threaten the influencer's authenticity and affect the audience's willingness to hear from the influencer (Audrezet, de Kerviler, and Moulard 2020; Boerman, Willemsen, and van der Aa 2017). Many users are drawn to influencers who are

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"ordinary" people who seem intrinsically motivated (i.e., without commercial interests) to provide useful information and thus are perceived as more authentic, relatable, and trustworthy than marketing communications (van Driel and Dumitrica 2021; Haenlein et al. 2020). In a market survey, 62% of consumers indicated that they were concerned that influencers capitalize on impressionable audiences and that they might stop following an influencer who posted sponsored content (Forrester 2018). Previous works on celebrity endorsement (Hung, Chan, and Tse 2011; Tripp, Jensen, and Carlson 1994) have found that celebrities who endorse too many brands or products might lose credibility with their audiences. Thus, compared with organic posting, brand sponsorship may hurt the influencer's reputation, leading to a *reputation-burning effect*.

On the other hand, several pieces of evidence support the prediction that, compared with posting organic content, brand sponsorship might help an influencer's reputation—a *reputation-strengthening effect*. Sponsorship is evidence of the influencer's worth and ability (Brooks 2020) and enables the audience to access exclusive brand information (Casaló, Flavian, and Ibáñez-Sánchez 2018). In our influencer video posting dataset (collected from YouTube),[3] influencers received more likes and fewer dislikes on average for sponsored content than for organic content. Furthermore, honesty is the best policy in word-of-mouth marketing (Abendroth and Heyman 2013), and the audience may feel that an influencer is honest upon seeing a brand sponsorship disclosure. The disclosure of sponsorship may help safeguard the audience's perception of authenticity, which is central to their relationship with the influencer (Evans, Wojdynski, and Hoy 2018; Mediakix 2020b).

In summary, the effect of brand sponsorship on an influencer's reputation remains ambiguous and empirically untested. This research fills the gap by asking the following questions:

1) What is the impact (main effect) of posting sponsored content relative to posting similar organic content on the influencer's reputation (operationalized as the number of subscribers)?

[3] https://youtube-dl.org/.

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Of course, users choose to subscribe and unsubscribe for diverse, multifactorial reasons.[4] We focused on brand sponsorship as one possible reason and controlled for other observable reasons and tested the sensitivity of our results to unobservable reasons (see Section 4.1.1).

We investigated three potential moderators: the influencer's audience size, the "fit" between the sponsored content and the influencer's usual content, and brand popularity.

2) Does the impact of brand sponsorship vary with the influencer's audience size? If so, what are the possible reasons?
3) Does the impact of brand sponsorship vary with a) the *fit* between the influencer and the sponsored content and b) the *popularity* of the brand?

We analyzed audience size because influencers with smaller audiences may generate more engagement and trust, and their sponsored content tends to be perceived as more authentic and trustworthy than the content of influencers with larger audiences (Barker 2019). Experticity (2016) found that 82% of consumers considered a purchase recommendation trustworthy if it came from an influencer with a relatively small audience.[5]

We analyzed "fit" because a high influencer–sponsorship *fit* is suggested to increase the effectiveness of influencer marketing messages (Avery and Israeli 2020), while a lack of *fit* can decrease the believability of the message and hurt the credibility and trustworthiness of the influencer. Similarly, the match-up hypothesis in the celebrity endorsement literature (Kirmani and Shiv 1998; Knoll and Matthes 2017) predicts that a better celebrity–brand *fit* leads to more-positive brand evaluations and more-effective celebrity endorsements.

[4] For example, subscribers may choose to stop following an influencer because of rumored or confirmed inappropriate behavior such as the old racist tweets of Elle Darby on YouTube. A viewer may follow or unfollow an influencer because of a friend's recommendation. A viewer may follow an influencer but then not watch any subsequent videos simply because the viewer stops spending time on YouTube. Such instances are unlikely to correlate with brand sponsorship and thus should not alter our main results.

[5] A "relatively small audience" is defined as 1,000 to 100,000 subscribers; these influencers may also be described as "micro-influencers," in contrast to "macro-influencers" (https://www.socialmediaexaminer.com/how-to-partner-with-youtube-micro-influencers/, accessed on January 11, 2022).

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Finally, we analyzed brand popularity because once an associative link is formed between an influencer and the brand, the audience's perceptions of the brand may spill over onto the influencer and vice versa (Batra and Homer 2004; McCracken 1989), just as endorsements from beloved celebrities can improve brand perceptions (Arsena, Silvera, and Pandelaere 2014; Miller and Allen 2012). In addition, the viewer's utility from viewing sponsored content may depend on the brand's popularity (Rochet and Tirole 2006), which may influence their willingness to (continue to) interact with the influencer who features a particular brand in creating the content.

This study analyzed a unique dataset of 85,669 user-generated YouTube videos created by 861 active English-speaking influencers in the beauty and lifestyle category. For each influencer, we collected the daily number of subscribers (as a proxy for the influencer's dynamic reputation) and daily content-posting behavior from August 2019 to July 2020. For each video, we collected the audience engagement information (the number of views, likes, dislikes, and comments), whether the video was sponsored or organic, and the video features according to four categories: basic video properties (Zhou et al. 2021), the influencer's emotions and appearance (Zhang et al. 2021), the influencer's voice (Hwang, Liu, and Srinivasan 2021), and visual aesthetics (Yeh et al. 2013). We included these categories because they can affect the viewer's attention, engagement, and perceived interpersonal relationship with the influencer (Ekman and Oster 1979; Zhang et al. 2022).

The treatment effect indicates the reputation impact of a YouTube influencer posting a *sponsored* video (the treatment group) compared with a counterfactual, when a similar influencer posts a similar *organic* video (the control group). To replicate such comparisons, we constructed comparable treatment and control groups by matching our samples at the influencer-video level. We employed the DiNardo-Fortin-Lemieux (DFL) reweighting method for the matching step (DiNardo, Fortin, and Lemieux 1996; details are in Section 3.2). On the basis of this matched sample, we report three main findings.

First, the empirical analysis of the matched sample revealed a *reputation-burning* effect: posting a sponsored video relative to posting an organic video with similar characteristics cost the influencers 0.19%

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of their subscribers on average. A back-of-the-envelope calculation suggests that the effect due to a single sponsored video translates to an average loss of roughly 79,800 views for an influencer in a year (the calculation details can be found in Section 4.1). *Second*, the *reputation-burning* effect was stronger among the influencers with larger audiences. To understand why, we analyzed three audience response variables: the number of likes, number of comments, and amount of trust reflected in the comment text. For all three variables, we found a larger gap between the sponsored and organic videos among the influencers with larger (vs. smaller) audiences. *Third*, an interesting heterogeneity was observed in the sponsorship effects. A high fit between the sponsored content and the influencer's "usual" content is likely to mitigate the reputation-burning effect, perhaps because a high fit makes the influencer look like a more credible source of product information (Sternthal, Dholakia, and Leavitt 1978). The reputation-burning effect was mitigated when the brand involved in the video was less well-known, perhaps because the influencer can provide brand information to which the audience would otherwise not have easy access (Casaló, Flavian, and Ibáñez-Sánchez 2018). Our main results were robust to an extensive set of robustness tests, including hypothetical unobserved confounders, alternative model specifications, and alternative mechanisms. Details are in Web Appendix Sections B, D, and E.

Our research effort contributes to the emerging literature on influencer marketing (e.g., Huang and Morozov 2022; Pei and Mayzlin 2021) by empirically testing a fundamental assumption of several theoretical works: that brand sponsorship causes reputation harm to social influencers (Fainmesser and Galeotti 2021; Mitchell 2021). For example, Mitchell (2021) modeled the dynamic relationship between an influencer and a subscriber. The relationship involves a "reap and sow" cycle in which the influencer oscillates between giving unbiased advice and monetizing the opportunity to advise. The derived cycle is built on the *assumption* that pursuing the most revenue (through sponsored content) will hurt the influencer's reputation, but we are among the first to *empirically* show the reputation-burning effect.

We also enrich the influencer marketing literature by focusing on how influencers' decisions to post sponsored content affect the outcomes for the influencers themselves. By contrast, prior research on the

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outcomes of influencer marketing has focused on the consumer and the firm rather than the influencers. Specifically, researchers have studied predictors of the number of likes and comments on sponsored contents (Ershov and Mitchell 2020; Hughes, Swaminathan, and Brooks 2019; Wies, Bleier, and Edeling 2022), the likelihood of reposting a sponsored content (Leung et al. 2022; Valsesia, Proserpio, and Nunes 2020), product sales (Bharadwaj et al. 2022), and purchase intentions (Lee and Eastin 2020). Leung et al. (2022) found that sponsored content was reposted more times when the influencer had more subscribers, when the post included the brand name in a hashtag, and when more of the influencer's subscribers had interests in the domain of the sponsored brand (such that the brand "fits" the subscribers). Ershov and Mitchell (2020) found that adopting advertising disclosure regulations leads to fewer likes and comments on influencers' content and lower consumer welfare, perhaps because followers in social media do not like sponsored content. In addition, our findings show the importance of studying outcomes for influencers as a complement to outcomes for firms. For example, although influencers with larger audiences are attractive candidates for brand sponsorship because they can reach many customers (e.g., number of reposts; see Leung et al. 2022), we found that for influencers with larger audiences, brand sponsorship has a more pronounced negative impact and elicits more adverse reactions from the audience. This is perhaps because influencers with smaller audiences often form stronger ties and build more trust with their audiences (Wies, Bleier, and Edeling 2022).

Our focus on influencer reputation contributes to the nascent literature on influencer video advertising. Rajaram and Manchanda (2020) implemented an interpretable deep learning architecture to analyze the relationship between influencer video content and video engagement. Yang, Zhang, and Zhang (2021) developed an algorithm to predict the sales lift from sponsored videos. We complemented these works by analyzing the video content itself to construct comparable sponsored and organic influencer-video pairs with which we can examine consequences for influencers rather than for consumers or firms.

Finally, our heterogeneity analyses complemented the existing research on how audiences react to influencer marketing, and the results provide interesting comparisons with the existing literature on

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celebrity endorsements (e.g., Amos, Holmes, and Strutton 2008). Consistent with the source credibility model and product matchup hypothesis in the celebrity endorsement literature (Choi and Rifon 2012; Sternthal, Dholakia, and Leavitt 1978), we found that influencer-video fit mitigates the reputation-burning effect. We speculated that sponsored videos may seem more authentic and trustworthy when on topics for which the influencer has built credibility and expertise. In contrast to the value transfer model (McCracken 1989), which implies that more popular products may spill over onto the influencer, we found that influencers fare better when they promote less-well-known brands.

Our findings have crucial managerial implications for influencers, brands, and social media platforms. For *influencers*, our estimation of the effect of posting a sponsored video offers a baseline with which influencers can evaluate the hidden economic trade-off involved in sponsorship offers. Our findings also suggest that influencers could be strategic in the timing and content of sponsored videos. It is advisable that influencers post sponsored content when the content is aligned with their usual style and when the sponsoring brands are less well-known. For *brands*, we offer insights into influencer selection; most brands currently use the number of subscribers as a primary criterion to choose influencers (Hwang, Liu, and Srinivasan 2021). However, our results suggest that influencers with more subscribers may not necessarily be the best brand partners. In fact, when promoting a brand, influencers with larger audiences tend to elicit more negative audience reactions, including fewer comments, lower engagement, and reduced trust, than influencers with smaller audiences. Such negative audience reactions may also extend to the promoted brand. Lastly, social media *platforms* should be advised that brand sponsorship will hurt influencer reputation and receptivity among audiences, potentially "turning audience away." Platforms could caution influencers about the risks of sponsorship and motivate them to post organic content, thereby reducing the exodus of active users who have lost trust in influencers and become less engaged on the platform.

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2. RESEARCH CONTEXT AND DATA

Our research context was YouTube, a major global influencer marketing platform. As of 2020, YouTube has an estimated 2.1 billion users. We focused on the beauty and lifestyle category because it attracts the highest volume of brand sponsorships (Schwemmer and Ziewiecki 2018).

YouTube influencers are often categorized by the number of subscribers: nano (1000 or less), micro (1,000–100,000), macro (100,000–1 million), and mega-influencers (more than 1 million) (Sonnenberg 2021). This breakdown reflects the assumption employed by many marketers that influencers' subscriber counts positively predict their marketing impacts. Our data include micro-, macro-, and mega-influencers. We do not include nano-influencers because they tend to post infrequently and are less likely to receive sponsorship offers (Weber 2018). We needed a sample of influencers who post sponsored content and post frequently enough that we can exploit variations in the postings. Following Hwang, Liu, and Srinivasan (2021), we collected data on 861 active influencers[6] in the beauty and style category on YouTube. We downloaded 85,669 videos (approximately 2.5 terabytes and 900,000 minutes) posted by the influencers between August 2019 and July 2020.

2.1. Influencer and Sponsorship Data

We obtained each influencer's number of subscribers and video views each day during the observation window from a data tracking website, noxinfluencer.com.[7] In the main analysis, we used the number of subscribers as the proxy for the influencer's reputation among their fans because gaining subscribers is a cumulative process similar to building reputation over time (Mitchell 2021). Intuitively, a user's decision

[6] We first retrieved the list of the top 1,000 most active influencers in the beauty and style category at https://channelcrawler.com/eng/results/1917 in August 2020 (the data displayed at the time of reading may differ from our dataset). We removed (a) invalid/unavailable YouTube channels, (b) videos that could not be found at the time of data retrieval, (c) videos with downloading errors, and (d) videos that resulted in feature extraction errors due to file corruption. We ended up with 861 influencers and 85,669 posted videos.

[7] An influencer's number of subscribers can change in two ways: new subscribers may follow the influencer, and old subscribers may unfollow the influencer. Both happen to every influencer, but we do not have data to differentiate between the two processes; instead, we observed the aggregate change in the number of subscribers over time. Our analyses uncovered the *relative* aggregate change between posting a sponsored video and posting a similar organic video.

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to follow or unfollow an influencer reflects the user's perception of their relationship with the influencer. We also collected key influencer characteristics, including their age, gender, and the length of channel title (measured by the character count). In addition, we documented the influencers' posting behaviors, including the number of days since their last video posting, the number of days since their first video posting, and their average video upload frequency.

In our study, we defined "a sponsored video" as a YouTube video created by an influencer in collaboration with a brand. We did not distinguish between the specific format of this sponsorship or how the brand was presented. Instead, following Hwang, Liu, and Srinivasan (2021), we identified sponsored videos on the basis of the influencer's sponsorship disclosure in the video description box (e.g., "This video is in collaboration with XX brand"). This approach aligns with the guidelines of the Federal Trade Commission on celebrity endorsements (FTC 2019), which mandate influencers to transparently disclose their brand collaborations. We found that 5,993 (7.0%) of the 85,669 videos were sponsored and that 333 (38.7%) of the 861 influencers posted at least one sponsored video during our observation window. We refer to the 333 influencers as "sponsored influencers" and the remaining 528 as "non-sponsored influencers."

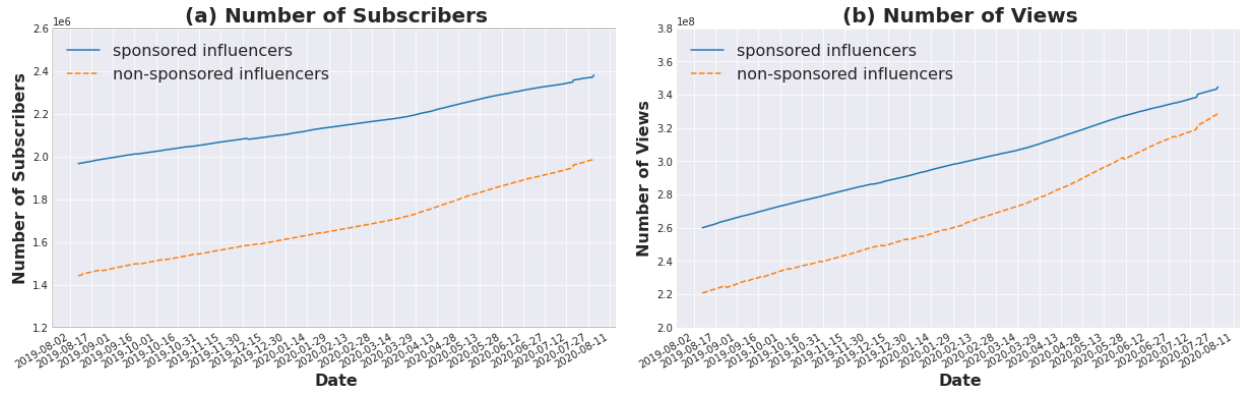


Figure 1. Reputations of Sponsored and Non-sponsored Influencers Over One Year

Note: "Sponsored influencers" are those who posted at least one sponsored video during our observation window, and "non-sponsored influencers" are those who never posted sponsored videos during our observation window.

Figure 1 illustrates the evolution of (a) the average subscriber count and (b) the average view count of the sponsored and non-sponsored influencers. Although the sponsored influencers had more views and

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subscribers than the non-sponsored influencers, the two groups converged slightly on both metrics during the observation window, consistent with the prediction that posting sponsored content may harm the influencer's reputation relative to posting organic content.

2.2 Video Data

For each YouTube video in the sample, we collected audience engagement information (the number of views, likes, dislikes, and comments) and video release information (date, video length, title, and description) from the video's YouTube page (see an example in Figure 2).

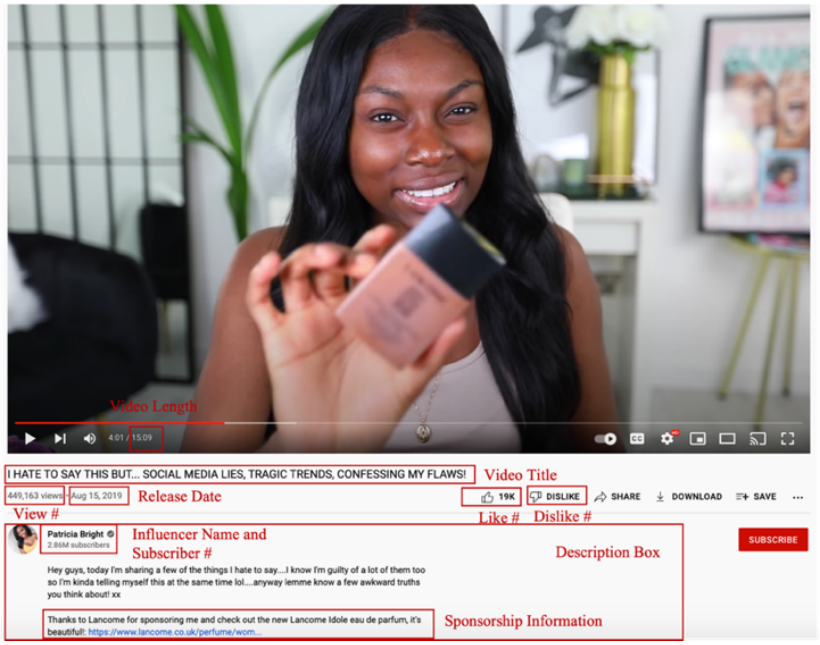


Figure 2. Information Collected from Each YouTube Video[8]

Note: This is a screenshot of a YouTube video posted by influencer Patricia Bright. The red boxes highlight the audience engagement information (the number of views, likes, dislikes, and comments) and release information (date, video length, title, and description). We extracted sponsorship disclosure information from the video description box.

We used machine learning models to extract the images, audio, and speech from each video. To capture what the audience can see, hear, and feel about the influencer, we then theorized and extracted 33 variables in four categories that may affect the viewer's attention and engagement: basic video properties and the influencer's emotions and appearance, voice, and visual aesthetics. Table 1 delineates the video features into four categories. The technical details are in Web Appendix Section A.1.

[8] https://www.youtube.com/watch?v=deudMVwDqqk, accessed on January 10, 2022.

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Basic Video Properties. We considered five variables (video length, scene number, average scene length, speaking rate, and sentiment) that may affect attention and engagement (Zhou et al. 2021).

Influencer's Emotions and Appearance. We first identified any human faces in the video, as the human face is a primary channel for nonverbal communication (Ekman and Oster 1979). For each video image (frame) that contains a face, we applied Face++[9] to quantify seven emotions and then averaged the values across the images for each emotion. We extracted five appearance features (gender, age, smile, attractiveness, and number of faces) that may affect the perceptions of the influencer's personality, which could affect their reputation (Zhang et al. 2021).

Influencer's Voice Features. An influencer may adjust their voice in sponsored videos (Hwang, Liu, and Srinivasan 2021). We used techniques from affective computing (Scherer, London, and Wolf 1973) and automated speech recognition (Eyben, Wöllmer, and Schuller 2010) to extract four vocal features (loudness, pitch, loudness variability, and talking duration) that may affect the viewer's perception of traits such as dominance (Scherer, London, and Wolf 1973), attractiveness (Fraccaro et al. 2011), and capability (Peterson, Cannito, and Brown 1995).

Visual Aesthetics. We measured two types of aesthetic features because aesthetics can affect viewer preferences and satisfaction (e.g., Moorthy, Obrador, and Oliver 2010; Yeh et al. 2013). Following Zhang et al. (2022) and Zhou et al. (2021), we extracted three motion features (foreground motion area, motion magnitude, and motion direction) and five color features (warm hue proportion, saturation, brightness, contrast of brightness, and clarity).

Table 1. Definitions of the Video Features

Basic Video Properties	
Video Length	Length in seconds.
Scene Number	Number of scenes.
Average Scene Length	Average amount of time between scene cuts (in seconds), computed based on intelligent scene cut detection.
Speaking Rate	Number of spoken words per second.
Sentiment	Average sentiment, based on the subtitle text.
The Influencer's Emotions and Appearance	

[9] https://www.faceplusplus.com/emotion-recognition.

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Emotion	We focus on five emotions: anger, happiness, neutrality, sadness, and surprise. For each emotion, we calculate a number in the range [0,100], with larger values indicating higher confidence in the state.
Gender	Gender analysis result; we transform it to binary (1 = female).
Age	Age analysis result; a non-negative integer.
Smile	A number in the range [0,100]; larger values indicating more smiling.
Facial Attractiveness	Separate values for female and male; a number in the range [0,100], with larger values indicating that the face is more attractive.
Face Sum	Number of faces detected in one scene.
The Influencer's Voice Features (Mean + Standard Deviation)	
Loudness	Vocal intensity as a narrow-band approximation from the voice signal.
Pitch	F0 (in Hz): the fundamental frequency.
Loudness Variation	Zero-crossing rate: the number of times in a given interval that the amplitude of the speech signal passes through zero.
Talking Duration	The fraction of time in a given interval that a person is speaking. Calculated as the inverse of the maximum frequency using the autocorrelation function (ACF).
Visual Aesthetics	
Motion Features	
Foreground Motion Area	Average percentage of the foreground in which motion occurs, computed based on background/foreground segmentation.
Motion Magnitude	Average motion magnitude measured in pixels, computed based on dense optical flow.
Motion Direction	Average motion direction measured in degrees, computed based on dense optical flow.
Color Features	
Warm Hue Proportion	Average proportion of pixels with warm colors (e.g., yellow, red) in a frame.
Saturation	Average saturation of all pixels in a frame.
Brightness	Average intensity of all pixels in a frame.
Contrast of Brightness	Standard deviation of the pixel intensity across the whole frame.
Clarity	The proportion of pixels with sufficient brightness (> 0.7) in a frame.

Note: For the measures that we estimated at the scene-level (the influencer's emotions and appearance features, and some basic video properties), we took the mean value of all scenes in the video to create video-level measures. As a robustness check, we used the emotion features calculated only in the first scene, and we obtained consistent findings.

3. EMPIRICAL FRAMEWORK

3.1. Treatment and Control Groups

The treatment, *brand sponsorship*, occurs whenever an influencer posts a sponsored video. We measured the impact of *brand sponsorship* on the influencer's reputation three days after the posting.[10] To

[10] We focused on the reputation three days after video posting for two reasons. First, most video views occur shortly after posting (a plot of the cumulative video views against the number of days since posting revealed a concave growth

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estimate the impact of brand sponsorship on the influencer's reputation, we needed to minimize the influence of other factors that systematically varied according to sponsorship status and that might affect reputation. Ideally, we hoped to find "twin influencers" (pairs with identical characteristics such as gender, reputation, and career stage) and randomly assign them "twin videos" (pairs identical in content, length, quality, etc., except for the sponsorship condition). However, this design is managerially infeasible. In Section 3.2, we describe how we constructed a matched sample to mimic or approximate such a design.

Our research context posed three challenges to the valid identification of the treatment effect from the observational data. First, brand sponsorship (i.e., the treatment) was irregular and sequential for each influencer: the influencers did not follow any particular timeline or pattern of video posting. Hence, we could not simply use one period or a uniform set of periods to split the data into pre-treatment and post-treatment periods. Second, the treatment effect might carry over to later periods such that a sponsored video might affect not only the influencer's immediate reputation but also the audience's reaction to subsequent videos. Third, the observational nature of the data comes with possible endogeneity from the influencer–brand selection process, leading to systematic differences between treatment and control groups. For example, brands may only work with certain types of influencers, and sponsored content may also be significantly different from organic content.

Inspired by Azoulay, Zivin, and Wang (2010), we addressed the first two challenges with our construction of the treatment and control groups of influencer-video pairs and addressed the third challenge with DFL reweighting in Section 3.2. By analyzing each influencer-video pair as a separate event, we eliminated consideration of within-influencer variation in posting patterns, thereby avoiding the issue of irregular, sequential video posting. An influencer-video pair belongs to the treatment group if the video is sponsored, and it belongs to the control group if the influencer did not post any sponsored videos during the observation window (i.e., non-sponsored influencers, as defined in Section 2.1). By excluding organic

curve). Second, three days is a short enough posttreatment window to minimize the risk of overlap between consecutive video postings (median number of videos posted per week in our dataset: 1.39). As a robustness check, we used the reputation two, four, and five days after video posting and obtained similar results.

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videos posted by sponsored influencers, we avoided the possibility of a carryover effect from a sponsored video to subsequent organic videos. The final sample contained 63,880 videos, with 57,991 organic videos in the control group and 5,889 sponsored videos in the treatment group.[11]

We followed the matched sampling approach employed in previous works that had similar research settings and identification challenges as our study (Azoulay, Zivin, and Wang 2010; Jäger 2016). For example, Azoulay, Zivin, and Wang (2010) estimated how the unexpected deaths of 112 academic "superstars" impacted their collaborators' publication rates. The authors constructed a matched control for each scientist who experienced the death of a superstar collaborator (i.e., superstar–collaborator pairs) and estimated the matched sample. Although our study is similar in many ways, the nature of an influencer's sponsorship decision was endogenous (as is the joint decision of the brand and influencer), while the premature deaths of collaborators were exogenous in Azoulay, Zivin, and Wang (2010). Given the treatment endogeneity in our setting, we matched organic influencer-video pairs and sponsored influencer-video pairs on a set of observed influencer and video characteristics that might influence the brand sponsorship decision. Furthermore, we employed the Rosenbaum bounds test to verify that our results are robust to the hypothetical unobservables that might affect the brand sponsorship process and influencer reputation simultaneously.

Table 2 presents the summary statistics for the treatment and control groups The groups were not balanced, as indicated by significant differences between the two groups. For example, the videos in the treatment group were longer (diff. = 290.8, $p \leq 0.001$) and contained more words (diff. = 116.4, $p \leq 0.001$). Such differences may influence both the likelihood and outcome of the treatment, thereby threatening the identification strategy (Athey and Imbens 2006). Next, we dealt with this issue by matching influencer-video pairs on their observed characteristics.

[11] The final number of video samples (63,880) is different from the number of raw video samples (85,669) because we excluded organic videos from sponsored influencers (i.e., those who posted one or more sponsored videos during the sample period). Audience reactions to these organic videos might have been influenced by preceding sponsored videos, and contamination might bias our estimates, similar to the challenges faced in Azoulay, Zivin, and Wang (2010) and Jäger (2016). In addition, we excluded the 104 sponsored influencer-video pairs that did not match.

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Table 2. Summary Statistics of the Key Variables

VARIABLES	(1) Treatment Group (5,889 videos)		(2) Control Group (57,991 videos)	
	Mean	**(Std. Dev.)**	**Mean**	**(Std. Dev.)**
Influencer's Reputation (in thousands)				
Channel_Preceding_Subscribers	2341.6	(2364.1)	1774.2	(1767.7)
Channel_Post_Subscribers	2348.5	(2371.1)	1788.5	(1774.4)
Influencer Characteristics				
Days_Since_First_Posting	2881.459	(1150.645)	1709.784	(1121.231)
Channel_Title_Length	12.0387	(13.9944)	15.1076	(34.7264)
Weekly_Video_Number	2.1095	(3.0906)	6.5447	(56.8605)
Age	27.7005	(151.5937)	17.9971	(328.4839)
Days_from_Last_Posting	5.9901	(80.4419)	2.7944	(37.9388)
Female	0.7818	(0.1706)	0.738	(0.1934)
Basic Video Properties				
Video_Length	857.6	(459.95)	566.8	(383.42)
Word_Count_Description	258.8	(152.08)	142.4	(133.94)
Scene_Number	92.37	(98.224)	91.14	(114.23)
Avg_Scene_Length	49.7	(166.97)	37.31	(132.12)
Sentiment	0.1616	(0.087)	0.08761	(0.1421)
Speaking_Rate	2.325	(5.432)	1.268	(9.6729)
Weekend	0.2445	(0.429)	0.2687	(0.4433)
Words_Length	1851	(1373.2)	437.9	(1100.3)
The Influencer's Voice				
Loud_Mean	0.8642	(0.317)	0.9558	(0.539)
Loud_Std	0.5271	(0.1573)	0.5454	(0.229)
Pitch_Mean	135.4	(32.107)	128.3	(47.349)
Pitch_Std	109	(15.884)	118	(26.35)
Verbosity_Mean	0.06453	(0.0223)	0.0852	(0.105)
Verbosity_Std	0.07552	(0.0357)	0.093	(0.0754)
Loudvar_Mean	0.6165	(0.0595)	0.5626	(0.1101)
Loudvar_Std	0.2099	(0.0273)	0.1884	(0.0502)
Visual Aesthetics				
Foreground_Motion_Area	0.3753	(0.2252)	0.4995	(0.2205)
Motion_Magnitude	11.35	(8.0535)	15.09	(15.022)
Motion_Direction	3.061	(0.2381)	3.047	(0.2780)
Warm_Hue_Proportion	0.3575	(0.1982)	0.4222	(0.2187)
Saturation	0.2689	(0.1350)	0.3359	(0.146)
Brightness	0.5764	(0.1628)	0.6156	(0.1519)
Cob	0.2251	(0.0513)	0.219	(0.0556)
Clarity	0.4242	(0.1483)	0.4292	(0.1488)
The Influencer's Emotions and Appearance				
Face (presence of face)	0.9063	(0.2914)	0.5573	(0.4967)
Face Count	1.824	(1.3097)	1.111	(1.355)
Smile	30.76	(22.127)	21.58	(27.706)
Facial Emotion				

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Anger	3.175	(5.4869)	2.278	(6.8636)
Happiness	25.46	(19.29)	16.5	(23.546)
Neutral	32.39	(21.177)	19.91	(25.343)
Sadness	8.828	(9.543)	6.184	(11.719)
Surprise	13.02	(12.061)	5.591	(11.679)
Image Quality of the Face	47.03	(25.025)	23.79	(29.604)
Facial Attractiveness (female rating)	58.1	(19.876)	35.29	(32.094)
Facial Attractiveness (male rating)	56.46	(19.327)	34.44	(31.344)
Facial Skin Status				
Health	39.48	(23.688	22.51	25.986
Stain	9.86	(8.3653	6.217	9.2708
Dark_Circle	9.313	(11.516	5.526	9.2443
Acne	5.781	6.3108	4.232	7.3766
Face Blur Condition	14.4	19.191	14.61	24.973

Note: We present the statistics for only the key variables owing to the page limit. *Channel_preceding_subscribers* is the influencer's reputation one period (three days) before posting a video, and *channel_post_subscribers* is the influencer's reputation one period (three days) after posting a video. All the statistical values were computed on the unweighted sample. The fewer observations (54,481) in the Emotion and Appearance category were due to some videos not containing a face. As a result, the video features in the Emotion and Appearance category could not be extracted.

3.2. Constructing a Balanced Sample Using the DFL Reweighting Strategy

Most sponsored posts are made through a two-step process: (1) the brand reaches out to the influencer with whom it hopes to partner, and (2) the influencer decides whether to collaborate with the brand. Brands choose influencers on the basis of several selection criteria such as reach/reputation (typically measured as the number of subscribers), engagement rate, relevance, content quality, and frequency of posts (Avery and Israeli 2020; Hwang, Liu, and Srinivasan 2021). For instance, the beauty tool and skincare product brand Vanity Planet considers both the influencer's subscriber count and average view count.[12] Influencers do not always accept sponsorship offers; they may decline because they want to take a break from making videos or are dissatisfied with a brand's financial offer (Hwang, Liu, and Srinivasan 2021; Mediakix 2020a). Thus, we should include influencer characteristics to account for the likelihood that an influencer will receive a sponsorship offer (e.g., days since the first video posting, the frequency of video postings, and pre-posting reputation) and the likelihood that the influencer will accept a sponsorship offer (e.g., whether the video is posted on a weekend, which might affect the influencer's availability).

[12] Vanity Planet influencer application website: https://www.vanityplanet.com/pages/influencer-application.

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Although we do not have data on factors such as the brand's marketing budget and the influencer's compensation, the audience also does not know such information. Thus, such factors cannot directly influence how the audience reacts to a sponsored video, although they may affect the audience's reaction *indirectly* through the content quality. For example, generous compensation may lead to a higher-quality video or a happier influencer in the video. We can capture these possible indirect effects by controlling for video features.

We used DFL reweighting (DiNardo, Fortin, and Lemieux 1996), a semiparametric weighting approach that balances a sample on observed characteristics. The DFL reweighting method constructs comparable treatment and control groups by first computing a sample weight for each given unit based on the propensity score (Rosenbaum 2002; Rosenbaum and Rubin 1985) and then assigning weights to each unit to eliminate any existing systematic differences between the treated and untreated units in terms of the observed covariates that explain the treatment probability. For example, if the control influencer-video pairs have a higher proportion of high-quality videos than the treatment influencer-video pairs, then the DFL procedure will downweight high-quality videos and upweight low-quality videos in the control group. That is, the DFL method creates a "synthetic" sample of weighted treatment and control groups in which the observed covariates are balanced across the two weighted groups (Zhang et al. 2021). The DFL reweighting method is widely applied in various contexts such as innovation (Bell et al. 2019), corporate finance (Yagan 2015), and labor economics (LaLumia 2013).

The DFL method serves a similar purpose as matching methods such as propensity score matching (PSM; see Rosenbaum and Rubin 1985). One significant advantage of the DFL method is that unlike matching methods, in which units in two groups that are not very similar to each other are discarded, the DFL method does not discard any unit when creating a balanced sample and thus better preserves the sample. This is because the "less similar" units are taken care of by the sample weight. Previous studies have also shown that reweighting methods such as DFL reweighting are more efficient estimates with lower mean

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squared errors than those from matching methods (see Hirano et al. 2003, Firpo 2007, Austin and Stuart 2015).

We employed the DFL method to match the sample at the influencer-video pair level and ensured that the weighted treatment and control groups were comparable on an extensive set of influencer and video characteristics. For the influencer characteristics, we included the frequency of weekly video postings, days since the last video posting, length of the channel title, number of days since the first video posting, age, gender, and the influencer's reputation three days prior to video posting. We included the video features as described in Table 1.

The DFL method was implemented as follows. For each influencer (*i*)-video (*j*) pair *ij*, we calculated the sample weight using the estimated propensity score, ps_{ij} (Rosenbaum and Rubin 1985), which is computed as a logistic function of the influencer and video features X_{ij} mentioned previously. We denoted the fraction of the sample in the treatment group as $\overline{sponsor}$. The sample weight with DFL reweighting for each unit *i* is given in the following equation:

$$w_{ij}(X_{ij},\hat{\beta}) = \begin{cases} 1 & \text{if influencer } i \text{ is in the treatment group} \\ \dfrac{ps_{ij}\widehat{(X_{ij}\beta)}}{1-ps_{ij}\widehat{(X_{ij}\beta)}}\left(\dfrac{1-\overline{sponsor}}{\overline{sponsor}}\right) & \text{if influencer } i \text{ is in the control group} \end{cases} \quad (1).$$

We validated the effectiveness of the DFL reweighting method by confirming that the weighted treatment and control groups were comparable on X_{ij}. We computed the standardized difference to assess the balance of covariates following the matching process (Rubin 2001, Austin 2011). This metric is computed as the difference between the means of the treatment group, $\bar{X}_{treatment}$, and the means of the control group, $\bar{X}_{control}$, normalized by the sample variance difference between the treatment group ($s^2_{treatment}$) and the control group ($s^2_{control}$). The smaller the standardized differences, the smaller the data imbalance. There is no strict cutoff in determining whether the weighted treatment and control groups are comparable enough, but as a rule of thumb, standardized differences of < 0.20 are generally considered very small, and those > 0.60 are considered large (Cohen 2013, McCaffrey et al. 2013). As detailed in Web

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Appendix Section D, across *all* covariates, DFL weighting effectively reduced the data imbalances, resulting in smaller standardized differences in the weighted sample than the raw sample. Notably, the absolute values of the standardized differences for all covariates are well < 0.2 (common criteria for "very small data imbalance"), with approximately 90% of covariates (i.e., 41/47 variables) in the weighted sample well < 0.1 (an indication of negligible data imbalance; see discussion in Austin and Stuart 2015). Hence, our weighting strategy created a matched sample that effectively mitigated the significant imbalances in the covariates that existed in the raw sample.

Lastly, the matching process dealt with an extensive set of *observed* factors as possible confounders. However, we could not exclude *unobservables* that might have influenced both the treatment status and the influencer's reputation simultaneously. To address this concern, we implemented a sensitivity test using the Rosenbaum bounds analysis. As discussed in Section 4.1.1 and Web Appendix Section E, the DFL-matched sample led to results that were fairly robust to hypothetical *unobservables*.

3.3. Estimator of the Reputation Impact of Brand Sponsorship

Equation (2) describes our main empirical specification:

$$Post_Reputation_{ij} = Constant + \beta \cdot Sponsorship_{ij} + \gamma \cdot Controls_{ij} + Month_{ij} + \varepsilon_{ij} \qquad (2),$$

where $Post_Reputation_{ij}$ is the reputation of influencer i one period (one period is three days) after posting video j. $Sponsorship_{ij}$ equals 1 if influencer i's video j is sponsored; otherwise, 0. The coefficient of interest is β, which represents the main effect of posting a sponsored video j relative to posting a similar organic video on the reputation of influencer i. We also controlled for an extensive set of factors in $Controls_{ij}$, including influencer i's characteristics (e.g., reputation three days *prior to* video posting, $Pre_Reputation_{ij}$) and video j's characteristics (e.g., the video characteristics used in the matching process and audience engagement variables). $Month_{ij}$ is the month fixed effect and captures the possible seasonal factors (e.g., audiences might be more active in summertime). Equation (2) does not include influencer fixed effects because they would be perfectly colinear with sponsorship status. As described in Section 3.1, in constructing the matched sample, we excluded organic videos posted by sponsored

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influencers from the matched sample. We additionally controlled for the influencer's posting behavior in Equation (2), including the mean and variance of the influencer's video characteristics over time and the number of days since the influencer's last posting. Lastly, we standardized all the video characteristics and log-transformed all count variables, including the reputation measure.

Equation (2) identifies the effects of brand sponsorship on influencer reputation through two differences. The first difference is the change in influencer reputation from before to after posting a *sponsored* video (d_1), while the second difference is the change in influencer reputation from before to after posting an *organic* video (d_0). The reputation change is captured in Equation (2), which used the reputation after a video posting ($Post_Reputation_{ij}$) as the dependent variable while controlling for the influencer's pre-posting reputation ($Pre_Reputation_{ij}$, which is included in $Controls_{ij}$).[13] The treatment effect of brand sponsorship on influencer reputation was then identified by computing the difference between the aforementioned differences, d_1 and d_0. In this identification, it is critical that d_1 is a counterfactual of d_0; that is, the videos and influencers in the two conditions are comparable. Comparability was achieved by the DFL matching process, where we constructed similar influencer-video pairs (see Section 3.2). In summary, we identified the treatment effect by assessing the relative change in reputation when an influencer posts a sponsored video as opposed to the counterfactual situation when a similar influencer posts a similar organic video.

4. *RESULTS*

4.1. Main Effect of Posting a Sponsored (vs. Organic) Video on Influencer Reputation

Table 3 reports the results of estimating our main model in Equation (2). In column (1), we estimate the model in the sample matched using DFL reweighting (as our main model); in column (2), we estimate the

[13] In Equation (2), the differences are implicitly captured by controlling for $Pre_Reputation_{ij}$. In Web Appendix Section B.1, we explicitly computed d_1 and d_0 by using the difference between $Post_Reputation_{ij}$ and $Pre_Reputation_{ij}$ for the same video as dependent variable. The alternative model specification yielded consistent results.

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model in the sample matched using PSM (as a robustness test). The results are similar between the two columns, which suggests that the findings are robust to the matching method. We focus on the results in column (1) for the rest of this section.

The estimated coefficient of the treatment status indicator, $Sponsorship_{ij}$, was negative and significant ($b = -0.00194$, $p < 0.001$), suggesting a *reputation-burning* effect, in which posting a sponsored video, as compared with posting an equivalent organic video (i.e., with similar video features), cost influencers 0.19% of their subscribers. Intuitively, as an influencer's fan base declines, views of their later videos will also decrease, holding everything else constant. Given that YouTube influencers make revenues based on their number of views, a back-of-the-envelope calculation suggests that the treatment effect of posting a single sponsored video translates into a loss of around 79,800 views, which is worth approximately $1,436.40 in annual income.[14]

Is the economic harm of the reputation loss balanced out by the commission fee for the sponsored content? The commission fee varies widely across platforms and by the number of subscribers.[15] For a YouTube influencer with approximately 1.18 million subscribers, which is the median of our sample, brands pay an average of approximately $23,600 per sponsored video. *Prima facie*, the revenue earned from the sponsorship appears to outweigh the cost of reputation harm. However, brands pay commissions as lump-sum payments, while the reputation harm caused by one sponsored post may have lasting effects. The average influencer in our sample had been active for more than six years as of July 2020, so the lifetime

[14] In our data, an influencer on average posts 150 videos a year. For a back-of-the-envelope calculation, consider an influencer with 1,180,000 subscribers (the median of our sample). The influencer receives 280,000 views on average for each video. Thus, assuming that reputation burning has the same effect on the video views, posting a sponsored video, as opposed to posting a similar organic video, would cause an influencer to lose 79,800 views annually (= 0.19% loss in views × 150 videos per year × 280,000 views per video). It follows that the annual loss of 79,800 viewership translates to 79,800 views per video × 0.018 USD per view = $1436.40. Note that YouTube influencers are paid according to their video views (through YouTube's payment and displayed ad), with the current rate being approximately $0.018 per view (https://influencermarketinghub.com/how-much-do-youtubers-make/).

[15] According to DigitalMarketing.org, Instagram influencers charge anywhere between $75 and $3,000 per sponsored Instagram post. YouTube influencers are usually more expensive, as they are paid $2,000 on average per 100,000 subscribers.

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effect of posting *one* sponsored post can accumulate to approximately \$8,618 (= \$1,436.40 × 6).[16] Though some other platforms may not compensate influencers like YouTube does, losing a number of subscribers is still costly for influencers because it will impact their ability to obtain profitable offers from brands in the future.

To counterbalance the *reputation-burning effect* of brand sponsorship, influencers may strive to enhance video quality while creating sponsored videos, but this is not an easy task. For example, the estimated coefficient of $Video_Length_{ij}$ is negative and significant ($b = -0.000294$, $p < 0.05$), which suggests that holding everything else constant, an influencer would need to create a video with a reduction of 6.6 standard deviation in length (0.00194/0.00294 = 6.6 standard deviations, as all video characteristics were standardized) to fully mitigate the negative effect of sponsorship on their reputation. This is not an easy task because the influencer would need to deliver the same effectiveness of content and messages in a much shorter timeframe. Similarly, the estimated coefficients of $Word_Count_Description_{ij}$ and $Saturation_{ij}$ imply that to bear out the harm of brand sponsorship on their reputation, an influencer would need to write 3.83 standard deviations (= 0.00194/0.000506) more content in video description or to have visual scenes that are 3.63 standard deviations (= 0.00194/0.000534) more vivid and colorful, holding everything else constant. Alternatively, influencers might take proactive actions to recover their reputations after posting sponsored videos, but these actions are costly on their own. For example, an influencer might send gifts to subscribers or post more frequent, high-quality organic videos (at the cost of scripting, filming, editing, and more). However, the same reputation-building actions could be taken by influencers who had never posted a sponsored video, and the benefit might be larger in this case.

To clarify, we do not claim that influencers should decline all brand sponsorships due to the *reputation-burning effect*. However, by quantifying the economic impact of reputation burning, we provide a baseline that influencers can use to think about the hidden economic trade-off involved in sponsorship

[16] We recognize that many influencers in our dataset continued to post after July 2020, so the average duration of a top-tier influencer's career is probably longer than six years. We adopted a conservative estimate of a YouTube influencer's lifetime by using six years.

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offers. Furthermore, influencers are appealing to brands precisely because they can engage a large audience. If an influencer cannot sustain or grow their fan base like other influencers, then the focal influencer may eventually lose sponsorship to a competing influencer (i.e., someone in the same category who has been able to grow their reputation). In addition, influencers are valuable to platforms because they engage large segments of the active user population. If many influencers are damaging their reputations via sponsorship, the platform might be worse off in the long run as overall audience engagement decreases.

Table 3. Main Effect of Posting a Sponsored (vs. Organic) Video on the Influencer's Reputation

VARIABLES	(1) Main Model (DFL reweighting)		(2) Robustness (PSM)	
	ESTIMATES	**S.E.**	**ESTIMATES**	**S.E.**
Sponsorship	**-0.00194*** **	**(0.000321)**	**-0.00164*** **	**(0.000327)**
	Influencer Characteristics			
Log_ Channel_ Preceding_Subscribers	0.996***	(0.000506)	0.995***	(0.000572)
Log_Days_Since_First_Posting	-0.00426***	(0.000430)	-0.00392***	(0.000401)
Log_Channel_Title_Length	0.00129*	(0.000534)	0.000436	(0.000646)
Log_Weekly_Video_Number	-0.000330	(0.000354)	-0.000335	(0.000481)
Log_Age	0.00119**	(0.000400)	0.000611*	(0.000311)
Log_Days_from_Last_Posting	-0.00166***	(0.000222)	-0.00171***	(0.000188)
Female	0.000224	(0.000411)	-0.0000404	(0.000485)
	Audience Engagement			
Log_Number_Views	0.00193***	(0.000331)	0.00160***	(0.000379)
Log_Number_Likes	-0.000114	(0.000300)	-0.000120	(0.000387)
Log_Number_Dislikes	-0.0000583	(0.000243)	0.000427	(0.000258)
Log_Number_Comments	0.000152	(0.000194)	0.00000772	(0.000268)
	Basic Video Properties			
Video_Length	-0.000294*	(0.000128)	-0.0000826	(0.000151)
Word_Count_Description	0.000506**	(0.000162)	0.0000949	(0.000162)
Scene_Number	-0.000274	(0.000195)	-0.0000809	(0.000250)
Avg_Scene_Length	0.000100	(0.0000794)	0.0000446	(0.0000719)
Sentiment	0.000344*	(0.000142)	0.000142	(0.000146)
Speaking_Rate	-0.000556***	(0.000159)	-0.0000254	(0.000169)
Weekend	0.000205	(0.000122)	0.000293	(0.000203)
Words_Length	0.000390***	(0.000107)	0.0000590	(0.000220)
	The Influencer's Voice			
Loud_Mean	0.000496	(0.000455)	0.000251	(0.000534)

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Loud_Std	-0.000490	(0.000492)	-0.000168	(0.000352)
Pitch_Mean	-0.00174**	(0.000563)	-0.000451	(0.000917)
Pitch_Std	0.00102**	(0.000385)	0.000688	(0.000543)
Verbosity_Mean	0.00110*	(0.000476)	0.000963	(0.000791)
Verbosity_Std	-0.0000436	(0.000311)	-0.000501	(0.000440)
Loudvar_Mean	0.00222**	(0.000706)	0.000599	(0.00116)
Loudvar_Std	-0.000448	(0.000327)	-0.0000505	(0.000330)
Visual Aesthetics				
Foreground_Motion_Area	-0.000325*	(0.000158)	0.000225	(0.000140)
Motion_Magnitude	-0.000190	(0.000167)	-0.000241	(0.000291)
Motion_Direction	0.000213	(0.000138)	0.000369	(0.000256)
Warm_Hue_Proportion	-0.000413*	(0.000176)	-0.000166	(0.000148)
Saturation	0.000534*	(0.000250)	0.000312	(0.000318)
Brightness	-0.000532	(0.000308)	0.0000596	(0.000188)
Cob	-0.000334*	(0.000132)	-0.000108	(0.000213)
Clarity	0.000133	(0.000282)	0.0000180	(0.000194)
The Influencer's Emotions and Appearance				
Face (Presence of Face)	-0.00354*	(0.00138)	-0.00317*	(0.00144)
Face Count	0.000444*	(0.000193)	0.0000355	(0.000294)
Smile	0.000215	(0.000205)	0.000383	(0.000232)
Facial Emotion				
Anger	-0.000241	(0.000188)	-0.000137	(0.000140)
Happiness	-0.000764*	(0.000388)	-0.000488	(0.000263)
Neutrality	-0.000699	(0.000413)	-0.000324	(0.000382)
Sadness	-0.000331	(0.000176)	-0.000258	(0.000207)
Surprise	-0.000292	(0.000216)	-0.000171	(0.000169)
Image Quality of the Face	-0.000384	(0.000236)	0.000267	(0.000186)
Facial Attractiveness (Female)	0.000875	(0.00224)	0.000286	(0.00154)
Facial Attractiveness (Male)	-0.000323	(0.00205)	0.000456	(0.00151)
Facial Skin Status				
Health	0.000422	(0.000281)	-0.000151	(0.000234)
Stain	-0.000107	(0.000204)	-0.00000873	(0.000170)
Dark_Circle	-0.000265	(0.000147)	0.0000418	(0.000102)
Acne	0.000112	(0.000111)	0.0000277	(0.000144)
Face Blur Condition	-0.000344	(0.000204)	0.000133	(0.000166)
Constant	0.0777***	(0.00676)	0.0877***	(0.00754)
Observations	63880		10732	
AIC	-855565.9		-63384.3	
BIC	-854387.5		-62430.5	

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Note: Column (1) shows estimates from the sample matched using DFL reweighting, and column (2) shows estimates from the sample matched using PSM. All video features were standardized, and count variables were log-transformed. *Channel_preceding_subscribers* is the influencer's reputation one period (three days) before posting a video (i.e., $Pre_Reputation_{ij}$). Fewer observations were made in the PSM sample because we used one-to-four matching and dropped all the unmatched observations. We also controlled for the mean and variance of the influencers' video characteristics but do not report them here owing to space limitations. For the full regression table, please refer to Web Appendix Section F. Standard errors are in parentheses and clustered at the influencer-month level. *p < 0.05; **p < 0.01; ***p < 0.001

4.1.1. Rosenbaum Bounds Analysis: Assessing Sensitivity to Unobservables

The identification of the treatment effect in Equation (2) was based on the sample that was matched on the observed characteristics. However, the DFL reweighting strategy, like all matching and weighting methods, cannot match samples based on unobserved characteristics. A potential concern arises if there are *unobservables* that affect both the decision to post sponsored content and the outcome of that decision, which would bias the estimated treatment effect. For example, if influencers tend to post sponsored videos when they are under financial stress, which affects the way influencers interact with their subscribers and worsens their reputations, then we would overestimate the treatment effect by misattributing the impact of financial stress on influencer reputation.

Although the presence of an *unobservable* is not empirically testable, we can assess the robustness of our main results to *hypothetical unobservables* by conducting a Rosenbaum bounds analysis (Rosenbaum 2002). In brief, consider two units, i and j, with observed covariates X_i and X_j and unobserved covariates U_i and U_j. The selection process (i.e., the probability of posting a sponsored video) is written as follows:

$$\begin{cases} Prob_i(Sponsorship = 1|X_i, U_i) = f(\beta X_i + \gamma U_i) \\ Prob_j(Sponsorship = 1|X_j, U_j) = f(\beta X_j + \gamma U_j)' \end{cases} \quad (3),$$

where γ indicates the influence of unobserved variables U on the treatment assignment process $Prob_i(Sponsorship = 1)$. Assuming a logistic functional form for $f(\cdot)$, we write the odds for each unit as $\frac{Prob_i}{1-Prob_i}$ and $\frac{Prob_j}{1-Prob_j}$. Then, the OR (odds ratio) of the treatment for the two units is:

$$\frac{odds_i}{odds_j} = \frac{\frac{Prob_i}{1-Prob_i}}{\frac{Prob_j}{1-Prob_j}} = \frac{exp\,(\beta X_i + \gamma U_i)}{exp\,(\beta X_j + \gamma U_j)} \quad (4).$$

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If the two units have the same observed covariates but differ on unobserved covariates (i.e., $X_i = X_j$ and $U_i \neq U_j$), the OR becomes:

$$\frac{odds_i}{odds_j} = \frac{\frac{Prob_i}{1-Prob_i}}{\frac{Prob_j}{1-Prob_j}} = exp(\gamma(U_i - U_j)) \quad (5).$$

Hence, both individual units have the same probability of posting a sponsored video (i.e., with the same propensity scores) if $U_i = U_j$ (i.e., the two units have identical *unobservables*) or if $\gamma = 0$ (i.e., the unobserved covariates have no influence on the likelihood of posting a sponsored video). In other words, unobserved factors do not introduce a hidden selection bias into our estimated effect if and only if the OR is 1. Otherwise, a potential bias may arise because the matching method would match units *i* and *j*, which have similar observables but differ in their actual propensity scores because of *unobservables*.

The Rosenbaum bounds analysis evaluates how inferences about treatment effects would change in the presence of $U_i - U_j \neq 0$, by varying values of γ and $U_i - U_j$ (and, hence, varying the OR). Following Rosenbaum (2002), we assumed for simplicity a binary unobservable variable: $U \in 0, 1$. Rosenbaum (2002; see details on page 107) showed that the OR is bounded by *Gamma*, which equals to $\exp(\gamma)$:

$$\frac{1}{Gamma} = \frac{1}{exp(\gamma)} < \frac{\frac{Prob_i}{1-Prob_i}}{\frac{Prob_j}{1-Prob_j}} < exp(\gamma) = Gamma \quad (6).$$

For Equation (6), Rosenbaum (2002; details are on page 110) developed a Wilcoxon signed-rank statistic *T* for the matched pairs $\{i, j\}$, where the outcome for the treatment is greater than the outcome for the control. According to Rosenbaum's (2002) proposition 13, for a given value of γ, if the treatment has a null effect, the unknown null distribution of *T* is bounded by two distributions of T^+ and T^-. Then, at each $exp(\gamma)$, we calculate a statistical significance level "p value" (both sig+ and sig−) and the 95% confidence interval based on the Wilcoxon signed-rank tests and the distributions of T^+ and T^-. Details on how to implement the Wilcoxon signed-rank tests are in Web Appendix Section E.

In Table 4, the column "Significance Level (sig+, sig-)" reports the estimated p values, and the column "95% Confidence Interval" reports the estimated 95% CI of the point estimate. Both columns report

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results when setting the influence of the hypothetical unobservable bias to $exp(\gamma)$, denoted as *Gamma*. We increased *Gamma* in steps of 0.05, starting from 1, and reported the estimated p values and intervals under each *Gamma*. A larger *Gamma* suggests a more robust result for hypothetical unobserved confounding factors, as it would require a larger influence of hypothetical *unobservables* to nullify the treatment effect.

Table 4. Sensitivity Analysis: Rosenbaum Bounds Test

Gamma (=$exp(\gamma)$)	Significance Level		95% Confidence Interval	
	sig+	sig-	CI+	CI-
1	0	0	-567000	-459958
1.05	0	0	-611375	-415166
1.1	0	0	-653625	-371908
1.15	0	0	-694184	-330031
1.2	0	0	-733346	-289525
1.25	0	0	-770842	-250185
1.3	0	0	-807116	-211596
1.35	0	1.40E-13	-842212	-173750
1.4	0	4.60E-10	-876114	-136708
1.45	0	3.10E-07	-909073	-100375
1.5	0	0.00005	-941056	-64583.3
1.55	0	0.002308	-971908	-29618.6
1.6	0	**0.03531**	**-1.00E+06**	**5187.47**
1.65	0	0.207489	-1.00E+06	39276.6
1.7	0	0.558735	-1.10E+06	72684.2

As our main analysis identified a (negative) *reputation-burning effect*, we are more concerned with a potential downward (negative) bias than an upward (positive) bias in the estimator. Thus, we examined the "sig-" and "95% Confidence Interval" columns. These columns lead to the same conclusion. The critical level of *Gamma* at which our *reputation-burning effect* might be nullified is 1.6, as this is where the 95% confidence interval first includes zero and corresponds to the turning point where we continue to obtain p < 0.05 from the Wilcoxon signed-rank tests. As *Gamma* increased to > 1.6, the p values would be > 0.05. In other words, the estimated negative treatment effect would be nullified if there were hypothetical *unobservables* (1) that affected the treatment assignment and influencer reputation at the same time and (2) that are so large that they could alter the *OR* by 1.6. Note that these results represent the worst-case scenarios

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and do not imply that such hypothetical *unobservables* exist and are large enough to nullify the treatment effect.

How good is the result of *Gamma* = 1.6? Our estimated *Gamma* value of 1.6 is at the upper end of the range reported in the literature (1.2–1.6; DiPrete and Gangl 2004; Manchanda, Packard, and Pattabhiramaiah 2015; Sun and Zhu 2013; Zhang et al. 2021), which implies that our results are on par with, if not more robust than, those reported in the previous studies. As the Rosenbaum bounds analysis provided a lower bound of confidence for the worst-case scenario, we are fairly confident that our results are robust to hypothetical unobserved confounders.

In addition, we followed Bharath et al. (2011) and provide an interpretation of the obtained value of *Gamma*. Specifically, we assessed how much change in the observed variables is required to change the *OR* by a factor of 1.60. We found that it would need an implausibly large amount of change in the observed video or influencer characteristics. For example, the influencer's reputation prior to video posting (*Channel_Preceding_Subscribers*) would need to increase by 2.42 standard deviations (translating to 4.45 million subscribers) to change the *OR* by a factor of 1.60. In Web Appendix Section E, we present the details of the Rosenbaum bounds analysis results and their interpretations.

4.1.2 Additional Robustness Tests and Ruling Out Alternative Explanations

Our results are robust to alternative model specifications, including using both the *relative* and *absolute* changes in influencer reputation as alternative dependent variables (see Web Appendix Section B.1). In a robustness analysis, we included organic videos of sponsored influencers and obtained similar results (see Web Appendix Section B.2). An analysis of views received by sponsored and organic videos suggests that brand sponsorship alone would not affect the influencer's ability to gain traffic; that is, we did not find evidence that YouTube's video recommendation algorithm explains the documented treatment effect by directing more traffic and thus more prospective subscribers to organic videos (see Web Appendix Section B.3). A series of analyses that controlled for brand and video topic fixed effects excluded the alternative explanation that the *reputation-burning effect* was driven by systematic differences in the characteristics of

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brands or topics featured in sponsored and organic videos (see Web Appendix Sections B.4 and B.5). In addition, we conducted two analyses where we did not control for video engagement or video characteristics and consistently obtained a negative and significant *reputation-burning effect* (see Web Appendix Section C.3). Finally, we conducted a placebo test and validated that the *reputation-burning effect* was attributed to the influencers' sponsored videos, rather than any potential disproportionate reputation growth between the treated and control influencers (Web Appendix Section B.6).

4.2. Heterogeneous Effect of the Influencer's Audience Size

Experticity (2016) and Barker (2019) showed that sponsored content is perceived as more authentic and trustworthy when posted by influencers with smaller audiences (i.e., fewer subscribers). We reason that the influencer's audience size might moderate the effect of sponsorship on reputation. We added an interaction term, $Pre_Reputation_{ij} \cdot Sponsorship_{ij}$, to the model in Equation (2), where $Pre_Reputation_{ij}$ is influencer i's (logged) audience size prior to posting video j. The coefficient of the interaction term, μ, captures the extent to which audience size moderates the treatment effect. In Table 5, μ is negative and significant ($b = -0.00125$, $p < 0.05$), indicating that the reputation-burning effect is stronger for influencers with larger audiences. That is, for every 10% increase in the influencer's audience size, the influencer's reputation experiences an *additional* decrease of 0.0125%, which is 6.32% of the average treatment effect. On the basis of our back-of-envelope calculation of the annual income loss from posting a single sponsored video, every 10% increase in audience size leads to an additional annual revenue loss of $91 (= 6.32% × $1,436.40).

Table 5. Heterogeneity by Influencer Audience Size

VARIABLES	ESTIMATES	S.E.
Sponsorship	0.0159*	(0.00745)
Sponsorship X Log_ Channel_Preceding_Subscribers	**-0.00125***	**(0.000507)**
Influencer Characteristics		
Log_ Channel_Preceding_Subscribers	0.996***	(0.000527)
Log_Days_Since_First_Posting	-0.00426***	(0.000430)
Log_Channel_Title_Length	0.00143**	(0.000522)
Log_Weekly_Video_Number	-0.000289	(0.000353)
Log_Days_from_Last_Posting	-0.00164***	(0.000223)

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Female	0.000218	(0.000408)
Log_Age	0.00117**	(0.000401)
Constant	0.0560***	(0.00650)
Observations	63880	
AIC	-855587.0	
BIC	-854399.5	

Note: The model was estimated on the matched sample using DFL reweighting. Our dependent variable was log-transformed. *Channel_preceding_subscribers* is the influencer's reputation one period (three days) before posting a video (i.e., $Pre_Reputation_{ij}$) and was also log-transformed. We controlled for video engagement and video features as in Table 3 and controlled for the mean and variance of influencers' video characteristics but do not report the coefficients owing to limited space. For the full regression table, please refer to Web Appendix Section F. Standard errors are in parentheses and clustered at the influencer-month level. *p < 0.05; **p < 0.01; ***p < 0.001

We reason that influencers with smaller audiences have tighter communities or network ties and form "a more intimate and meaningful relation" with their subscribers (Segura 2018; Wies, Bleier, and Edeling 2022: 12). If so, then smaller (vs. larger) audiences might be more receptive to sponsored content. For example, subscribers within a smaller audience might perceive a sponsored video as an achievement, whereas subscribers within a larger audience might perceive the same sponsored video as a brand endorsement, motivated primarily by monetary interests and therefore inauthentic and untrustworthy.

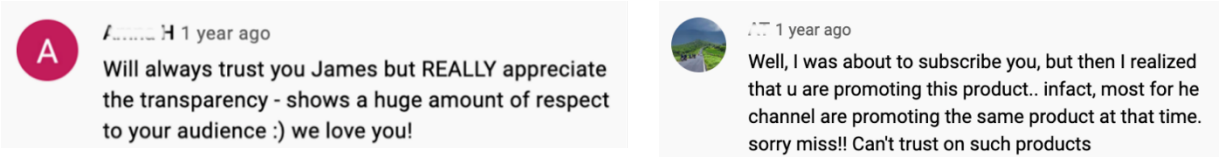
We tested this possibility by analyzing the audience's response to the post (reflected in the number of likes and comments; Ershov and Mitchell 2020; Rajaram and Manchanda 2020) and trust in the post and/or influencer (reflected in the comment text; Tabor 2020). An influencer's audience size may affect the variables of interest, so we operationalized them as rates (i.e., we divided the raw numbers of likes, raw numbers of comments, and trust measured by the number of subscribers), which is a common practice when measuring content performance (Keyhole 2020). We examined whether and how the audience response differs between sponsored and organic videos, and whether the *difference in responses* varies with the influencer's audience size.

We collected the number of likes, number of comments, and comment text for the 63,880 matched videos in our sample. Following Rajaram and Manchanda (2020), we analyzed the top 300 comments.[17]

[17] Some comments have replies from other audience members or from the influencer themselves; each reply counts as a separate record. The average video in our sample has 434.4 comments. The comments are ranked by YouTube's proprietary algorithm on the basis of timeliness and popularity (https://stackoverflow.com/questions/27781751/what-is-youtube-comment-system-sorting-ranking-algorithm).

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We found valid comments for 59,959 of the 63,880 matched videos in our sample,[18] yielding a total of 7,506,942 records. We used the Emotion Lexicon from the National Research Council (NRC; see Mohammad and Turney 2013) to analyze the emotions reflected in the comment text. The NRC Emotion Lexicon is a list of English words annotated by crowd-sourced workers. The lexicon includes the strength of the association between each word and eight basic emotions: anger, fear, anticipation, trust, surprise, sadness, joy, and disgust. We focused on *trust* because the trustworthiness of influencers is thought to be central to the effectiveness of influencer marketing and can plausibly vary with the influencer's audience size and be harmed by sponsorship (Audrezet, de Kerviler, and Moulard 2020; Tabor 2020). Figure 3 shows examples of comments that reflect high trust in column (1) and low trust in column (2). The names of audience members are blurred for privacy. For technical details and statistics on the trust measures, see Web Appendix Section A.2.



(1) (2)

Figure 3. Contrasting Examples of Comments on Sponsored Videos

We ranked the influencers by their subscriber counts and created three categories: small audience (bottom quartile), large audience (top quartile), and medium audience (middle two quartiles). Figure 4 visualizes the estimated effects of sponsorship on the three audience response variables for medium- and large-audience influencers relative to the baseline group of small-audience influencers. For all three variables, we found that the difference in audience response to the sponsored and organic videos was greater for medium- and large-audience influencers than for small-audience influencers;[19] the differences are slightly more pronounced for large-audience influencers than for medium-audience influencers. Figure 4

[18] Of the original 63,880 videos, 3,921 videos received no comments, disabled the comment function, or received only comments with emojis and scrambled characters.

[19] In the introduction, we compared the raw numbers of likes and comments between the sponsored and organic videos. We regressed the normalized numbers of likes and comments on sponsorship and the interactions between sponsorship and categorized audience size, influencer characteristics, and video features.

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supports our argument that the *reputation-burning effect* is stronger for influencers with larger audiences because larger audiences, relative to smaller ones, are less trusting and less engaged with sponsored videos than with similar organic videos. These findings can have consequential implications for brands that suggest that influencers with larger audiences may not always be the optimal choice in terms of favorable audience responses to brand-sponsored content.

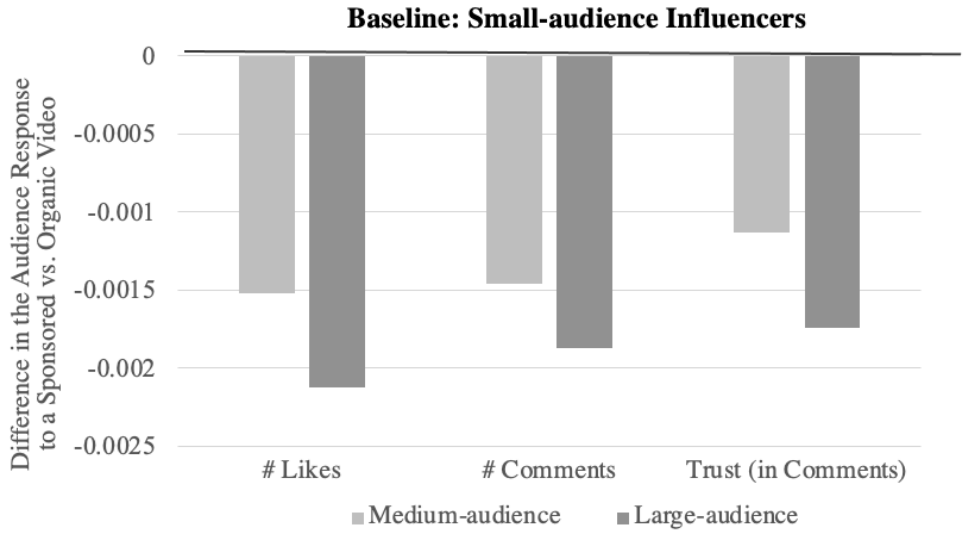


Figure 4. Influencer Audience Size Moderates the Audience Response to Sponsored Videos Compared with Organic Videos

4.3. Heterogeneous Effect by Influencer-Video Fit

We investigated *influencer-video fit* as another possible moderator of the reputation-burning effect because *fit* is important in celebrity endorsements (Kirmani and Shiv 1998; Knoll and Matthes 2017). Specifically, the "match-up hypothesis" states that a high celebrity-sponsorship *fit* leads to more-positive brand evaluations and stronger celebrity endorsement effects (Choi and Rifon 2012). For example, if an influencer usually shares her expertise in hair styling, then her subscribers might be more receptive to a sponsored video about a hair styling product (high fit) than to a sponsored video about a lipstick (low fit).

We use *doc2vec*, an unsupervised algorithm that learns fixed-length feature representations (i.e., "embeddings") from texts (Quoc and Mikolov 2014) in order to extract the content of each video from its

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title and description.[20] To characterize each influencer's "usual" content, we calculated the content embedding for each of their past videos, including both sponsored and organic videos, and we averaged them. To measure the fit between the influencer and the focal video, we calculated the cosine similarity between the focal video's content embedding and the influencer's average content embedding, regardless of the sponsorship status of the focal video. Technical details about the measures and examples are provided in Web Appendix Section A.3. Figure 5 shows the distribution of the *influencer-video fit* variable in our sample. The mean of the distribution was 0.69, and the standard deviation was 0.14.

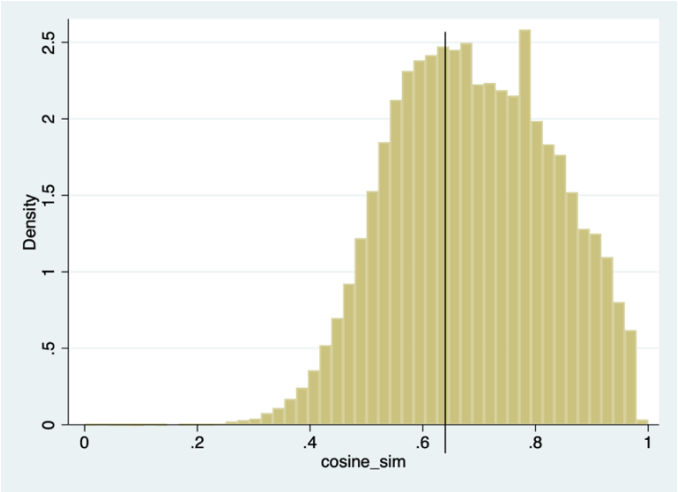


Figure 5. Distribution of *Influencer-Video Fit*
Note: The vertical line indicates where *influencer-video fit* is 0.63, 35th percentile of the influencer-video *fit* variable.

Equation (7) includes the linear and quadratic terms of $Similarity_{ij}$ to the model in Equation (2) to capture exploration and exploitation behaviors (He and Wong 2004; March 1991). In Equation (8), we added the interaction terms $Similarity_{ij} \cdot Sponsorship_{ij}$ and $Similarity_{ij}^2 \cdot Sponsorship_{ij}$. The coefficients of the interaction terms, μ_3 and μ_4, capture the extent to which *influencer-video fit* moderates the reputation-burning effect.

[20] The embedding based on the video title and description captures what the video is about, which is sufficient for our purposes. The embedding cannot speak to the presentation of the influencer and product; future studies could explore advanced algorithms to compute the embedding based on the raw video content.

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$$Post_Reputation_{ij} = Constant + \beta \cdot Sponsorship_{ij} + \mu_1 \cdot Similarity_{ij} + \mu_2 \cdot Similarity_{ij}^2 + \gamma \cdot Controls_{ij} + Month_{ij} + \varepsilon_{ij} \quad (7)$$

$$Post_Reputation_{ij} = Constant + \beta \cdot Sponsorship_{ij} + \mu_1 \cdot Similarity_{ij} + \mu_2 \cdot Similarity_{ij}^2 + \mu_3 \cdot Similarity_{ij} \cdot Sponsorship_{ij} + \mu_4 \cdot Similarity_{ij}^2 \cdot Sponsorship_{ij} + \gamma \cdot Controls_{ij} + Month_{ij} + \varepsilon_{ij} \quad (8)$$

In column (1) of Table 6, the coefficient of $Similarity_{ij}$ is positive and significant ($b = 0.0160$, $p < 0.05$), while the coefficient of $Similarity_{ij}^2$ is negative and significant ($b = -0.0132$, $p < 0.05$). The results suggest that influencers need some degree of consistent content but that over-consistency backfires. This aligns with the findings of past work on exploration-exploitation and variety-seeking behaviors (He and Wong 2004; March 1991). In column (2), the coefficients of the interaction terms ($\mu_3 = -0.0260$, $p < 0.05$; $\mu_4 = 0.0205$, $p < 0.05$) suggest a U-shape moderation effect of the similarity measure on influencer reputation. In particular, the vertex of the U-shape occurs at a $Similarity_{ij} = 0.63$, approximately representing the 35th percentile of the influencer-video *fit* variable (the 35th percentile is indicated by the vertical line in Figure 5). The results suggest that the moderation effect of influencer-video *fit* varies across videos (U-shaped relationship). For example, when the influencer-video *fit* is small (< 0.63), increasing the *fit* would result in a more negative reputation-burning effect, possibly due to customers' need for uniqueness (Abosag et al. 2020). However, once the influencer-video *fit* passes 0.63, the influencer-video *fit* variable positively moderates the reputation-burning effect, which is consistent with the "match-up hypothesis" (Kirmani and Shiv 1998). Approximately two-thirds of our sample are on the right side of the U-shape, where the influencer-video *fit* value is > 0.63. Hence, when promoting a brand, an influencer might consider creating content that *fits* their style.

As most of our sample is on the right side of the U-shaped curve, the findings suggest that influencers should strive for consistency overall while including enough variety so that the audience does not become bored (Chen et al. 2020). However, sponsored content is not a good way to provide variety. Rather, influencers appear to be better off (in terms of reputation) when sponsored content is consistent

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with their usual content (i.e., when $Similarity_{ij}$ is high). We reason that influencer-video *fit* mitigates the reputation-burning effect because the sponsored video may appear more authentic and trustworthy when on a topic for which the influencer has built credibility and expertise, consistent with the source credibility model (Sternthal, Dholakia, and Leavitt 1978). Arguably, *fit* depends not only on *what* is promoted but also on *how* it is promoted. For example, PepsiCo paid the pop singer Bebe Rexha to sample chips and then sync the flavors with her original music.[21] Although the product itself had a low fit with Rexha's usual content, the delivery of the promotion boosted the fit because she created an authentic analogy between creating music and developing new snack flavors.

Table 6. Heterogeneity by Influencer-Video Fit

	(1)		(2)	
VARIABLES	**ESTIMATES**	**S.E.**	**ESTIMATES**	**S.E.**
Sponsorship	-0.00198***	(0.000329)	0.00597	(0.00361)
Similarity	**0.0160***	**(0.00655)**	0.0176*	(0.00709)
Similarity^2	**-0.0132***	**(0.00528)**	-0.0145*	(0.00571)
Sponsorship X Similarity			**-0.0260***	**(0.0122)**
Sponsorship X Similarity^2			**0.0205***	**(0.00969)**
	Influencer Characteristics			
Log_ Channel_ Preceding_Subscribers	0.996***	(0.000522)	0.996***	(0.000525)
Log_Days_Since_First_Posting	-0.00429***	(0.000429)	-0.00429***	(0.000429)
Log_Channel_Title_Length	0.00140*	(0.000555)	0.00142*	(0.000560)
Log_Weekly_Video_Number	-0.000340	(0.000356)	-0.000342	(0.000356)
Log_Days_from_Last_Posting	-0.00163***	(0.000219)	-0.00163***	(0.000219)
Female	0.000225	(0.000410)	0.000231	(0.000410)
Log_Age	0.00116**	(0.000399)	0.00116**	(0.000399)
Constant	0.0743***	(0.00624)	0.0740***	(0.00620)
Observations	63880		63880	
AIC	-855603.0		-855603.6	
BIC	-854406.5		-854388.9	

Note: Column (1) shows the estimates from Equation (7), the model with the linear and quadratic terms of $Similarity_{ij}$. Column (2) shows the estimates from Equation (8), which additionally has the interaction terms $Similarity_{ij} \cdot Sponsorship_{ij}$ and $Similarity_{ij}^2 \cdot Sponsorship_{ij}$. Our dependent variable was log-transformed. *Channel_preceding_subscribers* is the influencer's reputation one period (three days) before posting a video (i.e., $Pre_Reputation_{ij}$) and was also log-transformed. We controlled for video engagement and video features, as in Table 3, and controlled for the mean and variance of influencers' video characteristics but do not report the coefficients owing to limited space. For the full regression table, please refer to Web Appendix Section F. Standard errors are in parentheses and clustered at the influencer-month level.

[21] https://www.warc.com/newsandopinion/news/pepsico-boosts-focus-on-influencer-marketing/43201.

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*p < 0.05; **p < 0.01; ***p < 0.001

4.4. Heterogeneous Effect of Brand Popularity

Finally, we examined brand popularity as a potential moderator of the *reputation-burning* effect because the value transfer model in celebrity endorsement research suggests that perceptions of the celebrity and brand may be transferable (Arsena, Silvera, and Pandelaere 2014; Batra and Homer 2004; McCracken 1989; Miller and Allen 2012). The popularity of a brand may influence how audiences react to sponsored content and how they interact with influencers because it affects the utility that audiences derive from viewing the content featuring the brand (Rochet and Tirole 2006). In the context of influencer marketing, an influencer might suffer less reputation harm from partnering with a highly popular brand (e.g., Lancôme) than with a less popular brand (e.g., Vanity Planet, a direct-to-consumer brand) because the popular brand might bring more utility to the audience, with a positive spillover effect on the influencer's reputation.

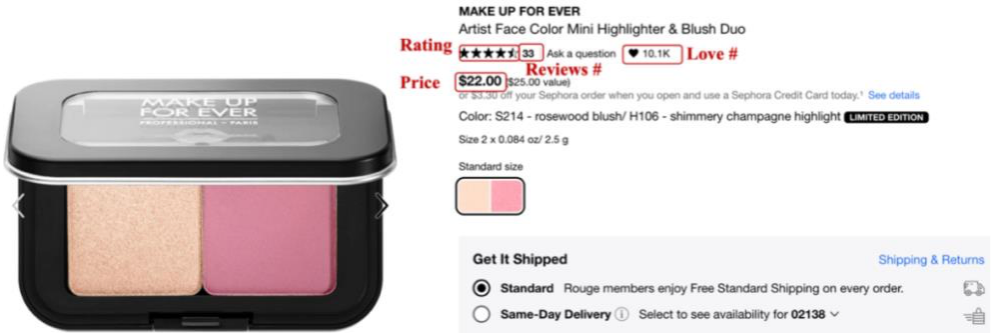


Figure 6. Product Information on the Sephora Website That Was Used to Calculate Brand Popularity

We collected the price, rating, review volume, and "love" count of each product on Sephora.com (see Figure 6), a leading multinational retailer of personal care and beauty products. For each brand in our sample, we averaged the information across all products offered by the brand. Then, we ranked the brands by their average "love" count (which reflects the desire of consumers to "save" the brand to their favorites list) and categorized the brands as *less popular* (the bottom 50%; less-well-known brands) and *more popular* (the top 50%; better-known brands). Some brands in our sample were new enough or niche enough

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that they did not appear on Sephora's website; we classified them as *non-Sephora*.[22] We controlled for the price, rating, and review volume to control for brand quality.

In Equation (9), we added the interaction term $Sponsorship_{ij} \cdot Brand_Popularity_{ij}$ to the model in Equation (2). $Brand_Popularity_{ij}$ is a three-class categorical variable that describes the popularity level of the brand in video j posted by influencer i: whether the brand is a popular brand ($More_Popular_{ij}$), a less popular brand ($Less_Popular_{ij}$), or is a relatively new or niche brand ($Non_Sephora_{ij}$). We set $More_Popular_{ij}$ as the reference group, so the coefficients of the interaction terms, ρ, capture the extent to which the *reputation-burning effect* differs when the brand is less-well-known or a non-Sephora brand.

$$Post_Reputation_{ij} = Constant + \beta \cdot Sponsorship_{ij} + \alpha \cdot Brand_Popularity_{ij} + \rho \cdot Sponsorship_{ij} \cdot Brand_Popularity_{ij} + \gamma\ Controls_{ij} + Month_{ij} + \varepsilon_{ij} \quad (9)$$

As shown in Table 7, the coefficient of the interaction term $Sponsorship_{ij} \cdot Non_Sephora_{ij}$ is positive and significant, and $Sponsorship_{ij} \cdot Less_Popular_{ij}$ is not significant at the 95% confidence interval but directionally positive. In other words, the *reputation-burning* effect is weaker if the sponsored video is for a less-well-known brand or for a niche/new brand than if it is for a better-known brand, contrary to the prediction of the value transfer model in celebrity endorsement research (McCracken 1989). This divergence may be attributable to the unique brand seeding process of influencer marketing, which often heavily influences subscribers' brand discovery (Carmicheal 2021). Casaló, Flavian, and Ibáñez-Sánchez (2018) found that audiences appreciate sponsored content for small and niche brands because they might otherwise not have easy access to the information, whereas information about popular brands is easy to find. It appears that influencers, more than celebrities, are expected to embody the informative role of advertising (Ozga 1960), and the audience derives more utility from viewing sponsored content when it contains

[22] We manually checked the sponsored brand list. A few large brands such as MAC, Ulta Beauty, and L'Oréal Paris are not listed on Sephora's website, so we assigned them the average value of all products on Sephora's website. The results are robust when excluding these brands.

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information about less-well-known brands (Rochet and Tirole 2006). See Web Appendix Section C.4 for supportive anecdotal evidence that audiences appreciate influencers who enable brand discovery, even in sponsored videos.

Table 7. Heterogeneity by Brand Popularity

VARIABLES	ESTIMATES	S.E.
Sponsorship (Reference: More_Popular)	-0.00309***	(0.000529)
Less_Popular	-0.000715	(0.000519)
Non_Sephora	-0.000289	(0.000404)
Sponsorship X Less_Popular	**0.000959**	**(0.000637)**
Sponsorship X Non_Sephora	**0.00127****	**(0.000479)**
Influencer Characteristics		
Log_ Channel_ Preceding_Subscribers	0.996***	(0.000511)
Log_Days_Since_First_Posting	-0.00426***	(0.000429)
Log_Channel_Title_Length	0.00133*	(0.000542)
Log_Weekly_Video_Number	-0.000345	(0.000355)
Log_Days_from_Last_Posting	-0.00165***	(0.000222)
Female	0.000252	(0.000415)
Log_Age	0.00125**	(0.000419)
Constant	0.0780***	(0.00696)
Observations	63880	
AIC	-855567.8	
BIC	-854353.1	

Note: The model was estimated on the matched sample using DFL reweighting. Our dependent variable was log-transformed. *Channel_preceding_subscribers* is the influencer's reputation one period (three days) before posting a video (i.e., $Pre_Reputation_{ij}$) and was also log-transformed. We controlled for video engagement and features, as in Table 3, and controlled for the mean and variance of influencers' video characteristics but do not report the coefficients owing to limited space. For the full regression table, please refer to Web Appendix Section F. Standard errors are in parentheses and clustered at the influencer-month level. *p < 0.05; **p < 0.01; ***p < 0.001

5. CONCLUSION AND DISCUSSION

We analyzed how an influencer's reputation is affected by posting a sponsored video relative to posting an equivalent organic video. We employed state-of-the-art video analytics on user-generated videos and used DFL reweighting to create matched treatment and control groups of influencer-video pairs. An empirical analysis of the matched sample revealed a *reputation-burning effect*; that is, posting a sponsored video,

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relative to posting a similar organic video, led to a loss of 0.19% of the influencer's reputation. The results were robust to hypothetical *unobservables* and extensive robustness tests.

We found that the *reputation-burning effect* was stronger among influencers with larger audiences. An analysis of audience responses, as reflected in the numbers of likes and comments, and trust in the comment text revealed a larger discrepancy in audience responses to sponsored and organic videos for influencers with larger (vs. smaller) audiences. We also explored heterogeneous effects by influencer-video fit and brand popularity. We found that a high *fit* between the sponsored content and the influencer's "usual" content mitigates the reputation-burning effect, perhaps because the high fit makes the influencer appear like a more credible source of product information (Sternthal, Dholakia, and Leavitt 1978). Furthermore, the *reputation-burning* effect is mitigated when the brand is less well-known, perhaps because the influencer can provide brand information to which the audience would otherwise not have easy access (Casaló, Flavian, and Ibáñez-Sánchez 2018).

Our study empirically validates the fundamental assumption of several theoretical works on social influencers that brand sponsorship harms influencer reputation (Fainmesser and Galeotti 2021; Mitchell 2021). We also contribute to the emerging research on influencer video advertising, the newest form of influencer marketing. Finally, we offer valuable comparisons with the celebrity endorsement literature (Choi and Rifon 2012; Sternthal, Dholakia, and Leavitt 1978). While celebrity endorsements might benefit more from collaborating with better-known brands, we found that the reputation-burning effect is mitigated when the sponsored post features less-well-known brands, highlighting the unique, informative role that audiences expect from social influencers.

This study has managerial implications for influencers, brands, and social media platforms. *For influencers*, our estimation of the effect of posting a sponsored video offers a baseline with which influencers can evaluate the hidden economic trade-off involved in sponsorship offers. Our findings also suggest that influencers could be strategic in the timing of posting and the content of sponsored videos. It is advisable that influencers post sponsored content aligned with their usual style (i.e., high fit) and whose

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sponsoring brands are less-well-known. Moreover, crafting high-quality sponsored videos such as videos with higher brightness, saturation, and motion magnitude level can effectively alleviate the reputation-burning effect (see Web Appendix Section C.1). For *brands*, our results may inform influencer selection. Most brands currently use influencer reputation as a primary criterion for choosing influencers (Hwang, Liu, and Srinivasan 2021). However, our results suggest that influencers with more subscribers may not necessarily be the best brand partners. In fact, when promoting a brand, influencers with larger audiences tend to elicit more negative audience reactions, including fewer comments, lower engagement, and reduced trust, than influencers with smaller audiences. Such negative audience reactions may also extend to the promoted brand. In addition, our results suggest that brands should recruit influencers whose usual content is closely related to the products to be promoted (i.e., high fit). *For platforms*, we caution that the increasing prevalence of influencer–brand collaboration may threaten long-term audience engagement. For example, Facebook announced the demise of Lasso, its short video-sharing platform (similar to TikTok), due to a lack of active users.[23] Platforms should be advised that brand sponsorship will hurt influencer reputation and receptivity among audiences, potentially "turning audience away." Platforms eager to mitigate this issue could consider charging influencers for posting sponsored videos, similar to advertisement monetization. Doing so could help offset the platform-wide audience engagement drop and discourage excessive sponsored content. Platforms could also caution influencers about the risks of sponsorship and motivate them to post organic content, thereby reducing the exodus of active users who have lost trust in influencers and become less engaged on the platform.

This paper is not without limitations. First, to ensure a sufficient sample size of organic and sponsored postings, we included only top and active influencers. Readers should keep this caveat in mind when considering the implications for nano-influencers. Second, we did not have access to YouTube's recommendation algorithm to fully examine whether it affects video views and, consequently, affects subsequent (un)subscriptions of influencers. In Web Appendix Section B.3, we found no significant

[23] https://techcrunch.com/2020/07/01/lasso-facebook-tiktok-shut-down/.

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difference in the video views to similar sponsored and organic videos, which suggests that the recommendation does not discriminate based on sponsorship status. We hope that future studies can gain access to the algorithm and explore this possibility more rigorously. In addition, we do not consider competition among influencers; for example, the growth of one influencer's reputation might affect another's ability to attract and retain followers. However, such a competition effect may be less profound in our context than in other contexts such as consumer purchase because audiences can follow many influencers simultaneously and do not have to unfollow one influencer to follow another. Lastly, we do not know the proportion of viewers who noticed each video's sponsorship disclosure. For example, if the disclosure appeared at the end of the description box, some viewers might not have seen it. Thus, our analyses report a conservative estimate of the reputation-burning effect, assuming that every audience is ensured to be aware of a sponsorship disclosure.

Our research motivates several directions for future research. First, to some extent, the *reputation-burning effect* implies a trade-off for influencers between sustaining their reputation by creating organic content and capitalizing on their existing reputations for profit through brand-sponsored content, even if it may come at the expense of their future reputations and potential profit opportunities. Thus, it is worth investigating influencers' optimal video-posting policy that maximizes their long-term payoff. Second, though our findings are arguably transferable to other contexts, future research should investigate the *reputation-burning effect* among influencers in other categories and on other platforms. Lastly, sponsored content can take place in various forms in terms of when and how a product is featured in the video. In this study, we do not differentiate these different forms, and we encourage future research to investigate further in this direction.

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